

Air University



Computer Networks (Lab)

Year 2025

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Department of Cyber Security

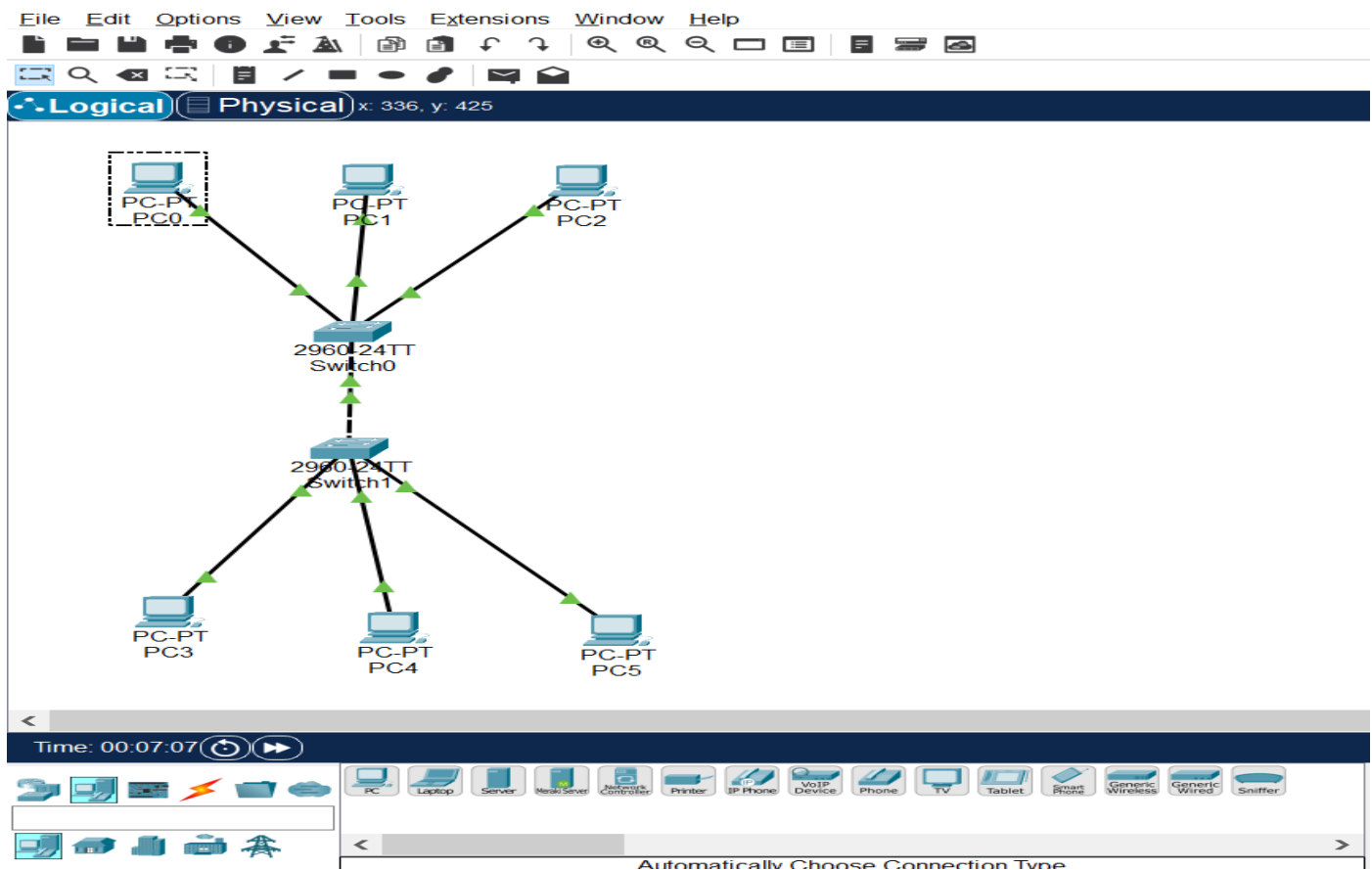
USAMA ARSHAD | 247141

BS – Cyber Security – Fall – 24 (A)

Date of Submission: October 09, 2025

Lab Task # 03 (A)

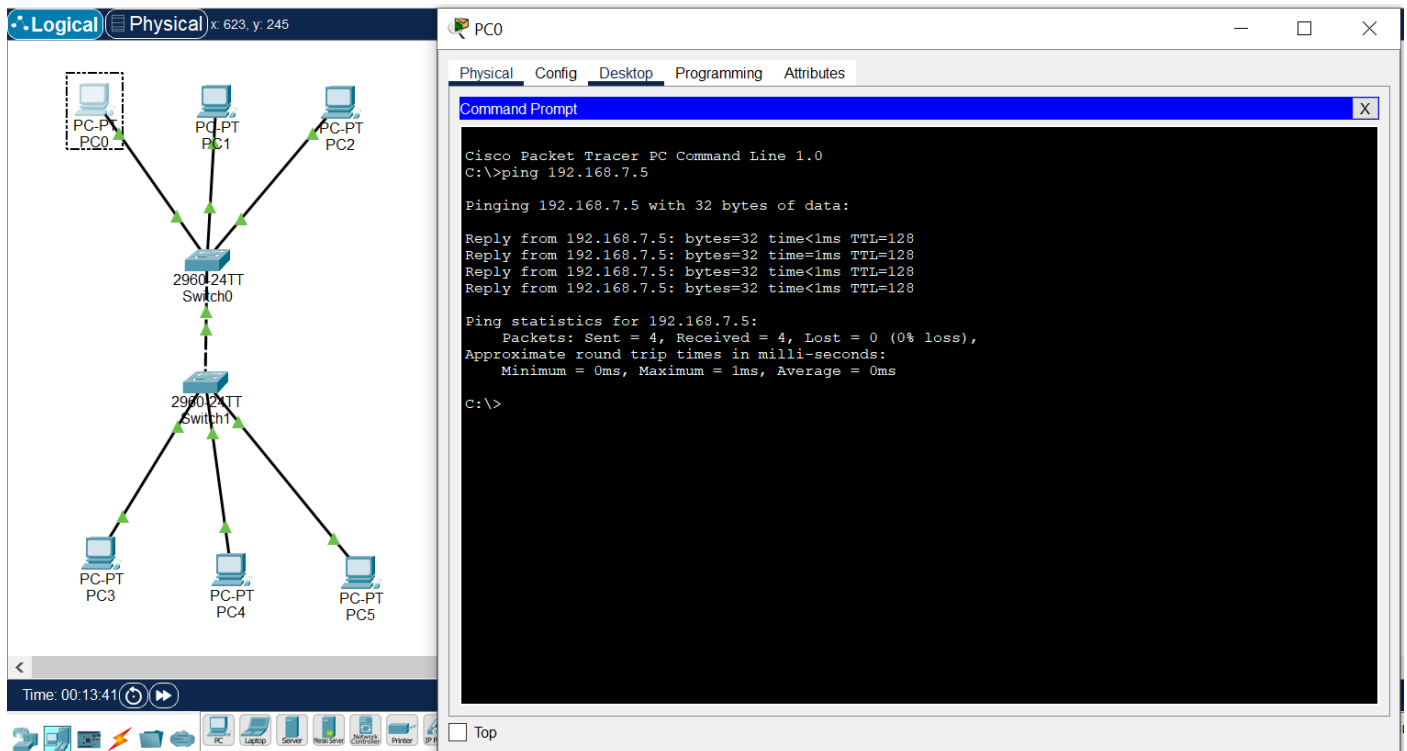
- Added required End devices and Network devices to implement given topology.



- Left click on each PC one by one, then go to Desktop and assign IP address.

The screenshot shows the configuration window for PC0 in Packet Tracer. The 'Desktop' tab is selected, and the 'IP Configuration' section is visible. The 'Static' radio button is selected for both IPv4 and IPv6. The IPv4 address is 192.168.7.1, Subnet Mask is 255.255.255.0, Default Gateway is 0.0.0.0, and DNS Server is 0.0.0.0. The IPv6 address is FE80::250:FFF:FEA3:750A, and the Link Local Address is FE80::250:FFF:FEA3:750A. The 802.1X section shows 'Use 802.1X Security' unchecked, Authentication set to MD5, and Username and Password fields empty.

- After assigning IP, Left click on any PC and open Command Prompt from Desktop. Use the command ping along with the IP of any other PC available in the network (e.g. ping 192.168.7.5)



The screenshot displays the Cisco Packet Tracer interface. On the left, a network topology is shown with two switches, 2960-24TT Switch0 and 2960-24TT Switch1, connected by a dashed line. Switch0 is connected to three PCs (PC0, PC1, PC2) and Switch1 is connected to three PCs (PC3, PC4, PC5). The right pane shows the configuration for PC0, with the 'Desktop' tab selected. A Command Prompt window is open, showing the output of a ping command to 192.168.7.5.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.7.5

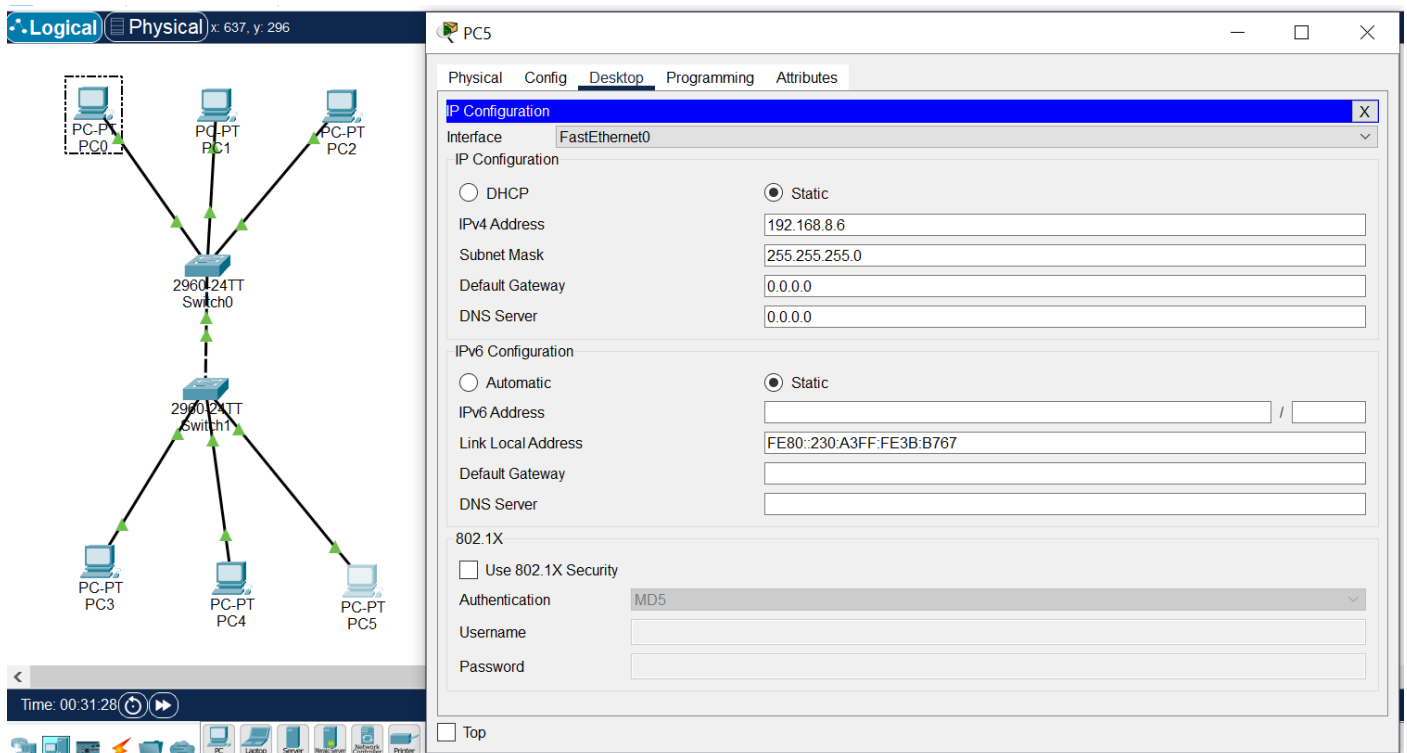
Pinging 192.168.7.5 with 32 bytes of data:

Reply from 192.168.7.5: bytes=32 time<1ms TTL=128
Reply from 192.168.7.5: bytes=32 time<1ms TTL=128
Reply from 192.168.7.5: bytes=32 time<1ms TTL=128
Reply from 192.168.7.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.7.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

- Changed the IP of PC-5 to 192.168.8.6 to check whether it stays in the network or not.



The screenshot shows the Cisco Packet Tracer interface with the same network topology. The right pane shows the configuration for PC5, with the 'Desktop' tab selected. The 'IP Configuration' window is open, showing the configuration for the FastEthernet0 interface. The IP address is set to 192.168.8.6, and the subnet mask is 255.255.255.0. The default gateway and DNS server are both set to 0.0.0.0. The IPv6 configuration is also shown, with the static IP address FE80::230:A3FF:FE3B:B767 and a link local address of FE80::230:A3FF:FE3B:B767. The 802.1X configuration is also visible, with the 'Use 802.1X Security' checkbox checked and the authentication method set to MD5.

IP Configuration

Interface: FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.8.6

Subnet Mask: 255.255.255.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80::230:A3FF:FE3B:B767

Default Gateway:

DNS Server:

802.1X

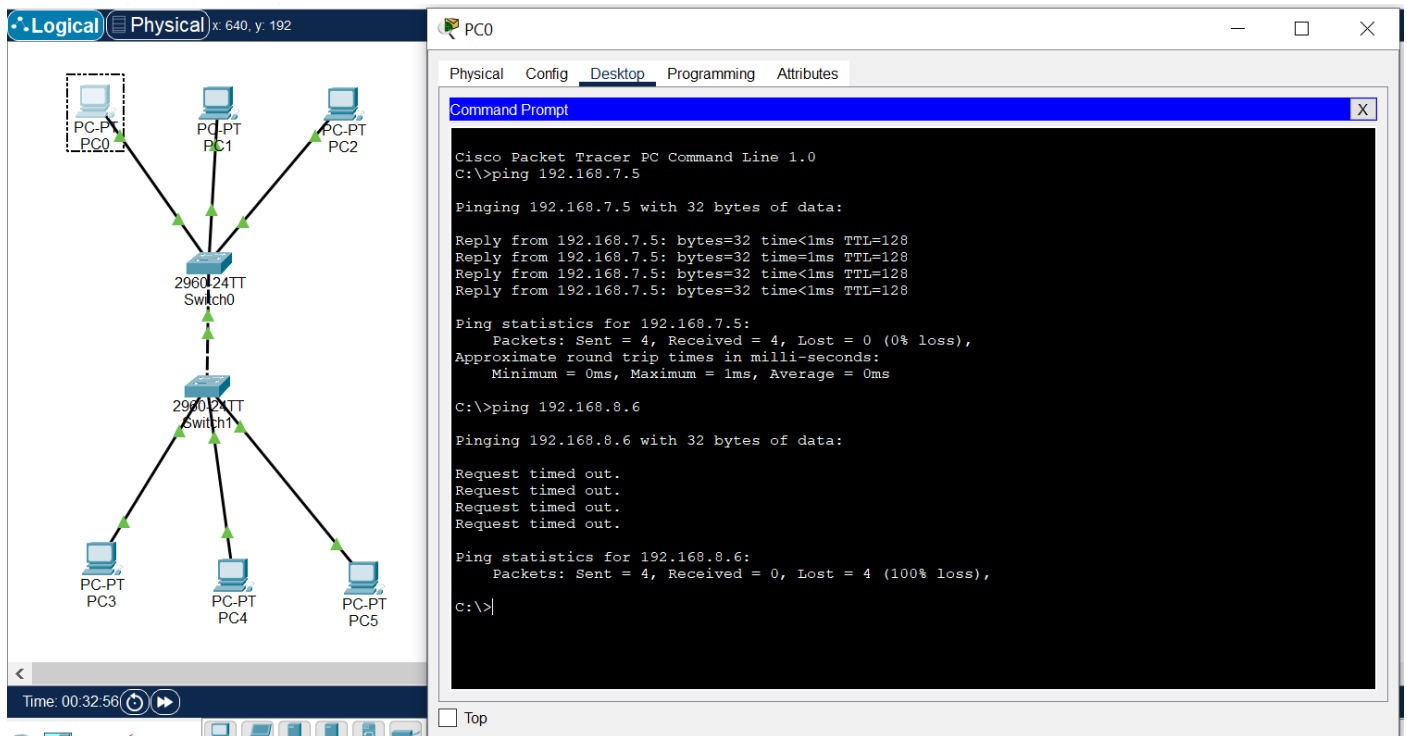
☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

- Checking the PC-5 from PC-0 using ping command, we can see that PC-5 does not exist in the network.



THE END